BAHIR DAR UNIVERSITY
BAHIR DAR INSTITUTE OF TECHNOLOGY
FACULTY OF: Computing
PAROGRAM: Information Technology

Course Guidebook for Network Design

1. GENERAL INFORMATION OF THE COURSE

Course Name:	Network Design
Course Code:	ITec4111
Course Credits Points:	5 hrs
Contact Hours:	2 hrs lecture & 3hrs lab
Course Module:	Network design, configuration and administration
Course Category:	Major
Status of Course:	Compulsory
Pre-Requisites:	Data Communication and Computer Networking(ITec2102)
Students' Department:	IT
Degree Program:	BSc
Students' Admission:	Regular
Year of Study: Semester:	4th : Second Semester
Academic Year:	2025/2026
Course Duration:	From February to June 2026
Course Chair:	Kokeb A.
Course Instructors:	Belayneh Teshome, MSc in Information Technology belayneh.teshome@bdu.edu.et Tesfahun Nurrie tesfahunnurrie16@gmail.com
Consultation:	Saturday morning 10:00AM -11:00AM

2. COURSE DESCRIPTION

This course is intended to teach students how to design and implement computer networks. The course covers detailed networking concepts like transmission media installation, switch and router selection and configuration, connecting to the internet, creating wired and wireless networks, implementing sub netting techniques and others. Students should be equipped with the latest networking technologies like Wi-Fi and how to design an efficient computer network.

3. OBJECTIVES OF THE COURSE

The main objective of the course is to introduce students with the fundamental concepts, standards and principles network design The specific objectives are:

1. Understanding how networks are integrated into business practices;
2. Design computer networks for enterprises
3. Deploy a small to medium sized networks
4. Understanding network simulation principles and methods;
5. Mastering the systems approach in network design;
6. Developing practical skills for network design in a heterogeneous environment.

4. COURSE LEARNING OUTCOMES (CLO)

At the end of the course, students will be able to:

No	Course Learning Outcome (CLO)	Performance Indicators (PIs)	Program Student Outcome (PSO)
1	Understand the basic concepts of network design	A1, A2, A3	1,2,3,4,5
2	Explain fundamental concepts and principles of network design	A4,A5	1,2,3,4,5
3	Understand different Techniques of and approaches of network design	A4,A5	3
4	Explain basic concepts of IP addressing and routing protocols	A1,A2,A3,A4,A5	3

5. THE COURSE TIME SCHEDULE AND TEACHING METHODLOGY

The following topics will be covered in this course within this semester.

Table 1. Time Schedule and Teaching Methodology of the course

Time Schedule	Course Contents	Teaching Methodology and References	Attaining student outcome
Week 1-3	1.1 Chapter 1: Applying a Methodology to Network Design • Existing network analysis • Identifying business goals • Technical goals and tradeoff	Lecture and tutorial Ref: 1&2	Outcome 1 & 2
Week 4-6	2.1. Chapter 2: Structuring and Modularizing the Network • Network design architecture • Types of network design architecture • Modular network • Network topology	Lecture and tutorial Ref: 1&2	Outcome 2
Week 7-8	2.2. Chapter 3: Designing Basic Campus and Data Center Networks • Types of network • Size of the network • Campus network • Enterprise network • Data center network • WAN and LAN technologies	Lecture and tutorial and lab Ref: 1&2	Outcome 3 & 4
Week 9-11	Chapter 4: Designing Remote Connectivity • What is remote connectivity • Why remote connectivity • Remote connectivity technologies • Virtual private network (VPN)	Lecture and tutorial and lab Ref: 1&2	Outcome 3
Week 12-13	• Chapter 5: Designing IP Addressing and selecting Routing Protocols • What is IP addressing • Types of IP addressing • Addressing Schemes • Routing protocols	Lecture and tutorial and lab Ref: 1&2	Outcome 3 & 4

Week 14-15	Chapter 6: Evaluating Security Solutions for the Network • Why network security • Network security fundamentals • Approach of network security • Network security protocols • Physical security • Access control	Lecture and tutorial and lab Ref: 1&2	Outcome 1,2, 3 & 4
	Chapter 7: Identifying Voice Networking and Wireless Networking Considerations • Telephone technologies • Traditional phones • VOIP • Wireless networking technologies • Wi-Fi • Cellular network	Lecture and tutorial and lab Ref: 1&2	Outcome 1,2,3 & 4
Week 16	Final Exam		

6. ASSESSMENT AND EVALUATION

The course utilizes a variety of continuous assessment and evaluation methods to ensure the successful achievement of course learning outcomes. These methods include assignments, projects, quizzes, midterm exams, final exams, classwork, presentations, and more. It is imperative that all assignments and tasks are completed within the specified deadlines.

Assignments and project: Assignments may involve collaborative work for brainstorming and developing general approaches. However, unless specified otherwise, each individual must complete their own submissions. Assignments and projects may entail both individual and group components, with equal participation expected from all group members. While collaboration is encouraged, submitting another individual's work as one's own constitutes academic dishonesty and will incur in penalties. All group members must contribute equally. Timely submission is mandatory; as late submissions will not be graded. Oral exams will accompany assignments and projects, requiring thorough preparation from all members. Additionally, the instructor reserves the right to assign additional tasks or assignments as necessary.

Quizzes: Quizzes serve to assess students' current understanding of the coursework. They are conducted in class following lecture or tutorial sessions.

Mid-term Exam and Final Exam: The exams will be closed book. During exam necessary materials will be provided by examiner.

Table 2. Summary of Assessments

No	Assessment type	Mark allotted	Content Coverage	Attaining course learning outcome (CLO)	Time Schedule
1	Assignment 1	10%	Chapter 2 and 3	1	Week 3
2	Assignment 2	10%	Chapter 6	2	Week 8
3	Mid exam	25%	Chapter 1,2 and 3	1,2,3,4,6	Week 8
4	Quiz	5%	Chapter 6		Week 10
5	Final Exam	50%	Chapter 4,5&6	1,2,3,4,5,6	Week 16
	Total	100%			

7. COURSE POLICY

Throughout the duration of this course, all students are expected to follow the student code of behavior as outlined in the "**Bahir Dar Institute of Technology Senate Mandate Legislation**" or "**BiT Senate Mandate Legislation, 2021"**. Academic misconduct, such as cheating, fabrication, and plagiarism, will not be permitted and will be quickly reported to the proper authorities for action.

Students who miss more than 15% of the semester's classes will not be eligible to take the final exam. Any student who surpasses the 15% absence level will receive a grade of 'NG' (No Grade) and must present valid reasons for their absence, according to Senate law, Art.37.2 and Art 37.3 (a) and (b) for NG management.

All students are required to have complete records of course assessments and evaluations. If a student misses a quiz, midterm exam, or any other continuous assessment without a valid reason, they will receive a score of zero. However, if a student fails to attend the final exam or complete a term paper/project without a valid reason, the instructor will record an "NG" according to senate legislation, Art. 40.8.

8. GRADING SYSTEM

The grading scale of the course is fixed scale according to senate legislation of Art. 177.

9. REFERENCES

1. NETWORK DESIGN REFERENCE MANUAL (NDRM)- 6TH EDITION
2. TOP DOWN NETWORK DESIGN (2ND EDITION) CISCO PRESS APPROVAL

The following bodies confirmed the course guidebook accordingly.

Position	Name	Signature	Date
Course Instructors	Belayneh Teshome		10/02/2026
	Tesfahun Nurrie		10/02/2026
Course Module Chair	Atalo H.		10/02/2026
Chair Holder	Kokeb A.		10/02/2026
V/Dean	Derejaw L.		